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The Clock Room and the Macro-Cognitive Accounting Framework—Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms: A Reader's Guide to Volume IV

Reading Guide Volume IV: Knowledge Production, the Discrete Leviathan, and a Brief History of Promises

Volume IV moves the question from how systems produce readings to how readings acquire knowledge, access, and eligibility for action. Academic publication, financial models, platform distribution, and intertemporal commitments all concern how judgment gains institutional recognition, and how that recognition in turn reshapes evidence, relations, and lives.

The eight chapters fall into four parts. Chapters 23 and 24 first analyze academic and financial knowledge production; Chapter 25 defines the Discrete Leviathan and sets the boundaries of historical comparison; Chapters 26–28 trace the platform history of "collecting you—understanding you—replacing you"; Chapters 29 and 30 then supply the economic base of the distribution-rights monopoly and take the promise as the center for discussing intertemporal obligation, re-judgment, and the bearing of responsibility.

I. Knowledge Production: How Readings Acquire Knowledge and Eligibility for Action

Chapter 23: A Brief History of Academic Knowledge Production—Publication, Evaluation, and the Growth of Knowledge

Journals, publication, impact factors, rankings, and professional evaluation organize knowledge production. Journals originally carried the functions of recording, circulation, and the confirmation of priority; they then formed new circuits with commercial publishing, professional evaluation, and digital concentration. Knowledge production thus occurs not only between researchers and their materials, but also between metrics and institutional qualifications.

The chapter first distinguishes promises, metrics, and judgment, and locates the three within the fiber framework. Metrics are not naturally neutral; they acquire meaning only within particular promises and judgment tasks. Changes in evaluation systems must be analyzed by returning to whether these tasks have been reorganized.

Metrics can help coordination, yet once they become targets they may change research behavior, so that publication growth in turn serves as proof of institutional success. The functional circuit may thereby turn into pathological recursion, and manipulation extends from content to signal. The text locates this closed loop as the presentation of four channels within one system: the metric rewrites the definition of the promise (a "good paper" becomes a "highly cited paper"), the direct corollary of dynamic corrosion acting on the promise side; fulfillment becomes metric attainment; meta-metrics calibrate old metrics; and cognitive debt accumulates. The Goodhart–Campbell tradition here receives its academic-scenario formulation—"The more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor"—and Strathern's restatement: "When a measure becomes a target, it ceases to be a good measure." Bornmann further conceptualizes the whole operation as "mimicry in science," with four adaptive strategies identified: following the mainstream of one's field and avoiding unconventional research; pursuing short-term research for faster publishable output; submitting to low-quality but bibliographically indexed journals; and slicing research findings into salami-thin pieces published separately.

The chapter also distinguishes disciplines, institutions, and reform paths, refusing to write the whole history of academia as continuous decline. Different fields differ in data, experimentation, theory, and evaluation cycles; reform cannot merely replace the name of a metric but must say what discriminative capacity has been added. Long-term support and slow science therefore belong to institutional conditions, not merely personal choices—in the text's words: "The value of long-term support lies in letting work that cannot yet prove itself through short-term publication continue to receive substantive examination"; while "an individual slowing down while still competing at the same publication threshold may merely concentrate the cost on the individual."

AI lowers the cost of text production but cannot supply research that was never conducted. Completion of text is not completion of research—the text distinguishes three kinds of acceleration: production acceleration reduces the labor needed to form a manuscript; dissemination acceleration makes manuscripts visible earlier and more widely; verification acceleration shortens, while preserving the force of examination, the process needed to exclude wrong explanations. A producer–receiver asymmetry follows: one side saves writing time while the other side's load of claims to process may grow. Open dissemination increases opportunities and changes verification

costs. Filtering and review can remove some problems, but cannot create, by post-processing, evidence that did not previously exist.

The chapter finally includes itself within the mapping operation: the delete test, the ten-item concept list, and its own terminology are all audited by the chapter's own standards. The self-reference does not weaken the argument; it requires readers to examine this chapter's evidence, case selection, and scope of conclusion by the same standards.

Chapter 24: The Evolution of Financial Knowledge Production—Judgment, Models, and Market Feedback

Financial knowledge exists not only in papers and formulas but also in data infrastructure, professional recognition, pricing tables, quotation systems, and trade execution. Portfolio selection, historical data, and evaluation benchmarks all carry their conditions of production; approximation error and prediction error must be kept apart. The text uses certification for a trackable action: researchers, clients, management, or regulators confer on a tool or user a corresponding degree of credibility and eligibility for use; whether certification overreaches depends on what the user claims the tool can accomplish—a history of adoption can show that a method was used, but not that all users drew the latter inference.

Risk adjustment, indexes, and printed pricing tables gave tools their eligibility for use; professional recognition and actual adoption do not always coincide. A common measure can support distributed judgment, but it can also hide model conditions inside seemingly neutral public readings. Historical data did not make judgment disappear; inputs always enter computation carrying their conditions of production. The distinctive difficulty of financial knowledge is therefore not merely that judgment is compressed into models and readings, but that the use of tools participates in forming the very market evidence later used to evaluate them—prices drawing closer to models, original return opportunities shrinking, others' positions providing reassurance, and collective execution conditions changing are mechanisms running in different directions; they cannot be merged into a single cyclical fable of "model success causing failure."

Once models are adopted, market evidence may be changed by the very use of the model. Reverse calculation can organize judgment but cannot generate independent evidence out of nothing—the text defines reverse analysis as asking, given an optimization model, risk estimates, and constraints, what expected returns would make the observed portfolio a reasonable choice; a conditional reverse computation, and "reverse calculation does not directly read investors' minds." Using an option price to back out a parameter and then plugging the same parameter back into the same formula to recover the original price only shows that reverse and forward computation are compatible; declaring the model independently validated on this basis reuses the same piece of evidence. Collective execution can record anomalies, but it may also bring market relations closer to the model, or shrink original opportunities through common adoption.

The history of portfolio insurance further shows that repairing the model and repairing the conditions of use are not the same thing. The text treats these changes as three interrelated kinds of work: the first is making data and computation more accurate, reducing detectable errors; the second is expanding the scope of the problem, bringing previously unrepresented execution, financing, and collective risks into judgment; the third is changing adoption conditions, so that new understanding can affect positions, scale, and responsibility, rather than appearing only in research reports—the three cannot substitute for one another. Local protection changes

collective execution conditions; known risks do not necessarily change adoption; divergences in execution ultimately produce different client outcomes. The production of information itself likewise requires incentives and procedures.

Judgment, calculation, and transaction records are not the same kind of knowledge; professional recognition, market fit, and actual risk understanding may also come apart. The growth of knowledge must be accounted for separately—judgment, tools, execution, and risk understanding; the adoption of some model cannot be written directly as knowledge having grown.

II. The Discrete Leviathan: Definition, Historical Comparison, and Platform History

Chapter 25: The Discrete Leviathan—Definition, Historical Comparison, and the Structure of Power

The Discrete Leviathan is defined as a cross-layer network of key nodes. The text's definition is worth quoting: a cross-layer network of key nodes formed by different organizations through technological dependency, property rights, and contracts, which—in a number of important knowledge, communication, or transaction activities—continuously stipulates the eligibility for participation, shapes what information is visible, and uses the results of activity to adjust the conditions of participation in the next round, while the individuals and organizations positioned downstream in the network face markedly asymmetric costs of substitution and exit. "Cross-layer" refers to the control linkages among infrastructure, content production, information distribution, transaction, and execution. "Discrete" denotes both that control and responsibility are distributed across different nodes and that activity is partitioned into units that can be recorded and processed; the two must not be conflated—the text also cautions that activity being discretely recorded does not by itself prove that cross-layer domination has taken shape.

Chained control, asymmetric substitution costs, cross-sovereign reach, and capital-structure reflexivity form the four discriminant dimensions, but the latter two are not absolute properties that every node must satisfy. The discriminant function identifies objects; it does not automatically yield conclusions about control intensity or consequences, still less a mathematically proven necessary-and-sufficient classifier. This form can also be called the loss function Leviathan—it requires no unified will, nor does it require all nodes to use the same loss function: individual nodes can pursue different and even conflicting objectives; what must be identified is how these objectives acquire external force through facilities, access, and distribution channels.

The search-and-advertising case concretely displays how crawling, indexing, ranking, ad exchanges, and reporting are connected, and how service benefits coexist with rule power. Whether a Discrete Leviathan is constituted still depends on the control linkages among different organizations and on the substitution paths actually available to dependents.

Universities, print, Purgatory, the East India Company, advertising, and behavioral design supply different historical mechanisms, not a single-line genealogy leading to the platform. The Purgatory material shows how a system of accountancy in which quantities of sin, quantities of suffrage, and lengths of punishment were converted into one another gave rise, in the hereafter, to a variable, measurable, and consequently manipulable time-scale; the East India Company material shows how, after the Company obtained the revenue-collection

rights over Bengal, Bihar, and Orissa in 1765 with collection backed by its own armies, a commercial organization acquired a capacity to rule that could directly alter the living conditions of others. The text's summary of these materials: "the historical materials above reveal different ways of organizing the transmission of knowledge, the maintenance of belief, the distribution of consequences, and the meaning of waiting; they did not form successive versions of one and the same machine, yet they show that human time and belief were not always already complete, prior to organizational relations." Structural similarity cannot substitute for evidence of historical continuity.

Common knowledge and fiber analysis further distinguish material exposure, mutual confirmation, preference change, and changes in the conditions of action. The definition also accepts counterexample constraint—in the text's formulation: "If an actor can migrate its activity at bearable cost, retaining the data, relationships, and business it needs, and the original node can no longer shape its conditions of participation, then the judgment of control ought to be narrowed accordingly"; if an institution lets independent evidence actually change the allocation of resources and the rules, the relevant correction mechanism cannot be described as wholly ineffective; if differences in distribution do not impede common records and coordination, one cannot, from personalization alone, declare the public observation interface to have collapsed. Local improvement likewise cannot be written as the disappearance of all problems.

Chapter 26: A Brief History of the Discrete Leviathan (Part I)—Collecting You (1994–2012)

From free services, viral spread, and behavioral surplus to the social graph and permanent online presence, this stage shows how users became transmission nodes, data sources, and conditions of value production. The free order lowered the threshold of entry and made the relation among advertising, data, and growth the institutional core.

Hotmail, Plaxo, and other transmission chains turned users into distribution devices. Hotmail appended a single line to the bottom of every email its users sent and signed up over 12 million subscribers in its first year and a half; it was the venture capitalists Tim Draper and Steve Jurvetson who in 1997 used the term "viral marketing" to describe this Hotmail practice. Plaxo formalized "virality" as a computable quantity—the new users each user could bring in depended on the number of update requests and their conversion rate; management computed the metric V at regular intervals to monitor the momentum of the viral engine. Like, Connect, and Open Graph connected relations and behavior to broader data collection: approval was discretized into a click, and the click was recycled as input to the ranking; even when a user had no account, "a separate set of data concerning individual browsing behavior can be created." Growth, collection, and capitalization gradually connected; behavior was no longer merely the result of use but became material that could be further processed.

The mechanisms have genuine service benefits and cannot be written a priori as mere manipulation. The same transmission and collection apparatus can serve public connection and can also serve prediction-for-sale; judgment depends on concrete use, feedback, and the configuration of rights. What must be traced is how service improvement turns into prediction-for-sale, how user labor is quantified, and how the social graph forms new structures of visibility.

Mobile permanent online presence changed the material preconditions of collection, extending behavioral records from fixed locations to everyday processes. The absence of governance also cannot be understood merely as regulators not seeing—in the text's condensation: the EU blocked by way of a rights framework,

the United States let things pass by way of market self-regulation and platform immunity, China substituted state capacity for individual protection—each path held its ground, yet none identified as an object requiring regulation the fact that extraction itself had already become the business model of free services. The failure of the formal layer has a quantitative footnote: for an American Internet user to read word for word the privacy policy of every website visited would take about 244 hours a year—consent thereby degenerated into a formality.

By 2012, the supply side, the user side, and the governance side had each been left with their own legacies—platforms had accumulated data assets, infrastructure, and an engineering pipeline for converting user behavior into extractable, optimizable signals; users' behavioral data, relationships, and cultural production had not yet been recognized as contributions carrying a claim to remuneration; what the governance side left behind was a structural gap. These laid the technical and economic foundation for the next stage, "understanding you." Subsequent governance faced not isolated products but a gradually consolidated growth structure.

Chapter 27: A Brief History of the Discrete Leviathan (Part II)—Understanding You (2012–2022)

The mobile Internet, algorithmic feeds, variable rewards, and the interest graph moved platforms from recording behavior to predicting and shaping it. Time order gave way to real-time response; the social graph was no longer the sole basis of distribution; platforms could match users with content directly.

TikTok's interest graph cold-starts on behavioral signals, reducing the mediation of social relations, and the distribution funnel changed creators' paths into visibility. In the text's formulation: "The weight of 'whom you know' in the distribution algorithm approaches zero; 'what you did' becomes the primary determinant"—the transformation thoroughly reconstructed the relation between creators and audience: their audience was no longer a community of active subscribers but anonymous traffic allocated predominantly by algorithm. Zhang Yiming's stated objective revealed a deeper ambition—"to eliminate the need for search across all corporate properties"—the platform's job is to push it to you before you know what you want. The recommendation system thus does not merely help users find content; it continuously stipulates what content is more easily produced and seen.

Formats of expression were thereupon reorganized by distribution metrics—from the internalization of constraints to the industrialization of hooks, to format compression and cognitive rewiring. The engineering deployment of variable rewards has a clear lineage: Eyal's four-step loop—Trigger → Action → Variable Reward → Investment—distinguished three kinds of variable rewards: rewards of the tribe, rewards of the hunt, and rewards of the self; former Google product manager Harris likened the smartphone directly to a slot machine and called competition among platforms "a race to the bottom of the brainstem"; his summary of its institutional mechanism was "Their business model is screen time, not life time." Variable rewards and engagement optimization affect not only single choices but may also change the intervals between choices and the shape of expression itself.

Governance responses and the evidentiary landscape changed at the same time. The text's summary of the net result of a decade of governance responses by the end of 2022: "a consent framework systematically bypassed by interface design (GDPR); a record fine requiring no change to product architecture (FTC, $5 billion); an internal-document leak that destroyed the 'we didn't know' defense yet produced no legally binding constraint

(Haugen); and a coroner's finding that established a causal chain for the first time yet produced no compensation (Molly Russell)." Formal consent and actual choice can come apart; the increase in internal documents, external whistle-blowing, and case evidence does not mean platform mechanisms have become transparent—they only change the conditions for obtaining evidence.

By 2022, there were both trained user responses and modes of production organizing expression around metrics. The question was no longer only what platforms know, but what platforms can arrange on the basis of that knowledge, and how such arrangement changes the content ecosystem and public governance.

Chapter 28: A Brief History of the Discrete Leviathan (Part III)—Replacing You and the Dual Diagnosis

Generative AI brought part of expression and action further into the system. RLHF, cloud labor, the three illusions of ground truth, the discretization of judgment and action, and the slide from cognitive offloading to will offloading form the continuous change from "understanding you" to "replacing you."

What counts as a true answer depends not only on whether the output has factual support, but also on annotation labor, data production, task definition, and who is qualified to confirm truth. The training engineering chain is condensed by the text as the "threefold zeroing"—"the starting point of verification zeroed (the system performs no causal checking, so any verification must rebuild the production process from zero), the material of verification zeroed (the chain of preference is compressed into a scalar, so the components of behavior cannot be decomposed), and the standard of verification zeroed (the coordinates of judgment are set unilaterally by the party being judged)." The cloud factory brings these conditions into the model while they may vanish at the interface of use: the three interlocking levels of labor control—intermediary, relational, and algorithmic—diffuse responsibility across the structure; the batch countdown, the ranking board, and coupon withdrawal variabilize wages and normalize penalties. The three illusions of ground truth press further on the value-theoretic premise: the so-called standard answer rests on a positivist error that "treats human disagreement as technical noise rather than a vital sociotechnical signal"; the HHH principle is in practice not culturally neutral; and in controlled experiments, RLHF-trained models did not become more correct but became better at persuading time-limited evaluators—false-positive rates rose by 24.1 percentage points on the question-answering task and 18.3 percentage points on the programming task; the text explicitly notes that the experiment's evidential weight is contested at the implementation level (a partial replication found the effect reversing after two fixes) and treats it as evidence that preference rewards can decouple from correctness, not as a settled characterization of RLHF pipelines generally. What remains—call it "alignment"—is not so much the model's approximation to human values as the model's fit to human evaluation behavior itself—precisely the middle link needed for the transition from "understanding you" to "replacing you."

From generation to execution, the system no longer only offers advice; it begins to act as an agent. The discretization of judgment and action means: "Judgment is sliced into an outsourceable stream of tokens (mapping elements), and action into delegable API calls." Cognitive offloading, judgment substitution, the automation of relational labor, and will offloading are not the same level; the content of authorization, stopping conditions, and the attribution of responsibility must be judged separately. The system can carry part of expression and execution, but the consequences of agency must still fall on an accountable subject.

Model collapse, pre-contaminated baselines, and frictionless machine-to-machine turbulence suggest that new

mapping chains may carry the accumulations of old systems at their starting point. The four-level regression also demands separate evidence—the text's ladder: the first step, cognitive offloading—users hand specific cognitive tasks to tools; the capacity to audit is present. The second step, attention hijacking—the algorithm decides what the person sees; the input to judgment has been manipulated. The third step, judgment substitution—the role regresses from "judge" to "confirmer." The fourth step, will offloading—"the obstacle shifts from capacity to willingness—one no longer cannot audit; one no longer wants to." The text also states explicitly that will offloading is a qualitative diagnostic tool, not a clinically validated category; one cannot presume, from the existence of assistive functions, that replacement at a higher level has occurred.

The dual diagnosis therefore points at both ends—training-side dismantling of the conditions of audit and use-side erosion of the will to audit—and the text specifies a two-front direction of treatment: preserve friction at the interface layer, install independent supervision at the institutional layer, protect independent audit in law, and allow audit conclusions to include abandonment. Who sets the goals, who authorizes, who can stop, and who ultimately bears the consequences remain questions that must continue to be tracked.

III. Distribution Rights and the Promise: From the Economic Base Back to Human Responsibility

Chapter 29: The Economic Base of the Discrete Leviathan's Distribution-Rights Monopoly

Chapter 29 supplies the economic structure for this network. Advertising is not merely the transmission of information; it also contains the contest for attention and the contest for distribution rights. The historical materials of Johnson, Norris, Laird, and Marchand anchor the century-long history—Johnson had already observed in 1759 that advertisements are "so numerous that they are very negligently perused" and that regaining attention requires "magnificence of promises," leaving behind the sentence: "Promise, large promise, is the soul of an advertisement." Together with data on advertising shares, circulation peaks, and digital subscriptions, the decline of newspapers is rewritten as the replacement of the attention-aggregation function: weekday circulation peaked in 1984, while the advertising revenue peak came only in 2005—this fifteen-to-twenty-year gap shows that what supported ad prices was the newspaper's attention aggregation position in its local press market, not total audience; once Google handled search advertising, Craigslist handled classifieds, and Facebook handled display advertising, what newspapers lost was not their audience but the "aggregation function."

Section 230 and Section 512 supply conditional limitations of liability—in the text's formulation: the combined effect is that platforms can carry user-produced content free of charge, with the legal risk of carrying user content at scale significantly reduced—"conditional limitations, not comprehensive immunity." Recommendation systems supply the apparatus of algorithmic distribution, and the concentration of advertising revenue supplies the economic conditions of continuous transformation. Platforms can thereby connect user content, behavioral data, and targeted auctions, turning advertising from general exposure into the condition of admission to a specific consumer's attention—what traditional media sold was "blind-box" exposure, buying the full page without knowing who saw it; what platforms sell is "admission tickets into a specific consumer's attention."

Attention admission tickets may also change the entry conditions of downstream product markets: who can reach consumers, and who is filtered out before entry. Prat and Valletti define digital platforms as "attention brokers": when the concentration of attention brokers rises, they have an incentive to create an "attention bottleneck" by

reducing the supply of targeted advertising—once the bottleneck tightens, more visible positions are occupied by incumbents, and entrants are filtered out before consumers become aware of their products. The concentration of the advertising market therefore affects not only platform revenue; it may rewrite the competitive structure of downstream retail markets along with it.

Expression compression, content farms, creator discipline, and gigification show how the producer end was transformed by the same mechanism. Traffic and monetization logics changed the value of content, while platform paternalism combined visibility rules with moral authority. Traditional media attempted to buy back distribution rights through vertical integration and content acquisition, repeatedly proving that content assets do not equal distribution control.

The chapter finally forms the causal chain: "advertising revenue concentrates → distribution rights become a monopoly → both ends are transformed." The common basis of both transformations is that the concentration of advertising revenue gives the platform the structural condition for transforming the two ends simultaneously—for consumers, the advertising monopoly means limitless extraction of behavioral data; for producers, it means visibility becomes a resource that the platform can allocate and withdraw unilaterally. Behavior collected and preferences modeled on the consumer end form a symmetric structure with expression compressed and visibility allocated on the producer end. The "zero-click" news consumption of AI conversation further heralds the possible upgrade of distribution rights into generative rights: what publishers face is not traffic shifting but traffic evaporating—no longer merely how much content is allowed to be seen, but whether original content still needs to be seen at all.

Chapter 30: A History of the Promise

The promise brings actions that have not yet occurred into the present, and binds performance, interpretation, compensation, and re-judgment into relation. Promise, prediction, persuasion, evaluation, and reliance must be kept apart—in the text's formulation: "A prediction makes a judgment about whether something will happen; a promise makes some action or guarantee an item for which the promisor is answerable. Persuasion attempts to change others' views or actions, without necessarily taking on any future obligation"; an evaluation offers a judgment about an object, and does not by that token become a guarantee of the object's future performance; reliance concerns how others act on it. Fried's formulation supplies the coordinate: "A promise invokes trust in my future actions, not merely in my present sincerity."

The Eastern and Western lineages supply different promise apparatuses and the right of reserved judgment. Trust cannot by itself determine the legitimacy of action; present revocation cannot automatically cancel a prior choice. Between keeping the promise and re-judgment, continued examination of conditions, consequences, and the affected must be preserved.

Once the promise enters institutions, the externalization of enforcement does not automatically mean the disappearance of responsibility. Interpretation is itself a normative activity; performance, compensation, and restoration are not the same thing. Measurable risk can be recorded; the unmeasurable remainder cannot thereby be deleted—Knight distinguished measurable from unmeasurable uncertainty: the former designates situations in which the distribution of outcomes can be known, transferable through grouping and insurance; the latter designates situations too particular to be grouped, which cannot be eliminated by pooling more similar cases. The text draws the exact corollary: "the metric records some effect as a measurable reading, while the substantial

part of the promise falls precisely outside the reading. Each expansion of the apparatus redraws the boundary between the measurable and the unmeasurable; the place of judgment is at that boundary."

Metrics in the academic, financial, and platform scenes acquire meaning only when returned to their questions: academic metrics answer whether research responsibility is verifiable; financial metrics correspond to prediction, contract, or fiduciary obligation; platform metrics may merge promise, evaluation, and the power of interpretation into the same infrastructure. The same metric does not imply the same structure of responsibility—the text's judgment: "Here the metric makes an obligation that was vague easier to verify. Problems arise when the metric is asked to answer what lies beyond what it records"; a record of an expression's effect or of an action's outcome cannot, without further explanation, become proof that all the promises have been kept.

After AI enters, the three-party arrangement in companionship relations, conversational personas, and agent authorization further change the position of the promise. The structural tension of companion products is summarized as "cruel companionship"—users seek relief from loneliness while maintaining their autonomy, whereas developers are structurally motivated by the imperative to maximize engagement and monetize interactions; "a theoretical diagnosis of particular product arrangements, not an incidence finding that all use of AI produces the same outcome." The agent, by contrast, is arranged to complete choices and actions on the user's behalf—"consider a shopping agent: the user allows it to compare goods within a budget and a purpose and to complete purchases." The system can act on behalf of the user; authorization must still have specific content; a conversational persona cannot replace the subject of responsibility. Exit cannot automatically restore the prior state; re-judgment also needs a position from which to take effect. Keeping the promise cannot exempt one from explanation, and re-judgment cannot exempt one from bearing the consequences.

IV. Interfaces between Volume IV and the Other Volumes

Volume IV advances the problem of Volume I. Fibers analyze the interpretations not yet excluded by knowledge readings; MCAF tracks the intertemporal consequences of metrics and institutions; and the psycho-social equilibrium reminds us that knowledge production and promising both occur within concrete relations. Together they prevent metrics from being treated as a naturally neutral form of knowledge.

Volume IV extends the application of Volume II. AI is no longer only a model producing credible appearances; it becomes the infrastructure of knowledge production, platform distribution, and agent execution. The completion of text, the completion of research, the growth of knowledge, and the completion of responsibility are thereby separated, and mapping chains enter further into academic, financial, and platform institutions.

Volume IV mutually attests institutions with Volume III. Model adoption and market feedback in financial knowledge production connect with the manifestation and governance of financial cognitive debt. The adoption of a model is not the growth of knowledge; the recognition of a metric is not the completion of judgment; historical governance experience cannot replace independent analysis of the new distribution structure.

Volume IV sets governance tasks for Volume V. The definition and counterexamples of the Discrete Leviathan, distribution rights and generative rights, and the promise and the subject of responsibility together require governance to address not only output quality but also the positions of entry, authorization, execution, payout, and re-judgment. The algorithmic central bank and public audit must therefore answer: who makes the rules, who bears the consequences, and who can still change the arrangement.